

National Institute of Technology Patna

Mid Semester Exam, May 11, 2022

Subject Code: PH24101

Time Allotted: 2 hours

Subject Title: Engineering Physics

Total Marks: 30

Attempt all questions

Q.1 Consider the damped forced harmonic oscillator equation

$$m\ddot{x} + \nu\dot{x} + kx = F_0 \cos \omega t$$

where $m = 0.2 \text{ kg}$, $\nu = 4 \text{ Nm}^{-1}\text{sec}$, $k = 80 \text{ Nm}^{-1}$, $F_0 = 2 \text{ N}$, $\omega = 30 \text{ sec}^{-1}$

(a) For the steady state solution $x(t) = A \cos(\omega t - \delta)$ find A and δ .

(b) Find the energy dissipated in one cycle.

(c) Now consider the damped forced harmonic oscillator equation above, but with the forcing as $F_0 \sin \omega t$. Solve it for A . (Do not make use of above mentioned numerical values)

[4+4+4]

Q.2 (a) Clearly explain, with suitable mathematical expressions, the concept of the quality factor Q in the context of a damped oscillator.

(b) A free damped oscillator of mass $m = 0.1 \text{ kg}$ is attached to a spring of $k = 0.9 \text{ Nm}^{-1}$ and is moving in a viscous medium. Its average energy decays to $1/e$ of its initial value in 4 sec. Estimate the damping coefficient ν and the quality factor Q .

(c) Set up the charge equation for an LCR circuit and interpret L , C and R in the context of an unforced damped oscillator. Obtain the quality factor Q .

[4+4+4]

Q.2 A height function is given as

$$h(x, y) = \frac{100}{2x^2 + 3y^2 + 2}$$

(a) Find $\vec{\nabla} h$. Plot some contour lines in the xy plane and indicate the gradient vectors on them.

(b) Use the gradient vector to find the slope of the tangent to the contour lines at an arbitrary point.

[3+3]